

Builder Journal — Week 1

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Why did you choose Computer Vision?

I chose Computer Vision purely out of personal interest — it is a field I have always been drawn to. What draws me to it most is how closely it mirrors the real world around us: almost every CV task, from recognizing an object to tracking motion, maps directly onto something we do naturally with our own eyes every day. Seeing an algorithm replicate that kind of understanding from raw pixels genuinely impressed me the first time I saw it work, and that initial impression is what pulled me toward pursuing it seriously. It feels less like an abstract branch of AI and more like teaching a machine a skill humans already rely on constantly, which makes the progress immediately visible and motivating to work on.

What applications interest you most?

I am most interested in applications built around object detection and tracking, in almost any form they take. Whether it is identifying objects in a static frame or following them consistently across a video stream, this category of application feels like the foundation that most real-world Computer Vision systems are built on — from security and surveillance, to retail and logistics, to sports analytics and autonomous systems. I am drawn to the breadth of that space rather than one narrow use case, and I am looking forward to exploring more of these applications hands-on as the fellowship progresses.

What did you learn this week?

This week gave me a full-stack view of a CV project, not just the algorithms. I learned how the modern Computer Vision pipeline actually works end-to-end: acquisition, preprocessing, detection, classification, segmentation, tracking, and deployment — and practiced explaining each stage clearly enough to write a technical report on it. On the engineering side, I set up a professional GitHub profile and repository, and built the Vision Toolkit application twice using two different architectures: once as a Streamlit web app, and once as a native desktop app first with PyQt5, then rebuilt again with an HTML/CSS/JavaScript frontend connected to a Python/OpenCV backend through pywebview. Implementing the same features (grayscale, Canny edge detection, blur filters, thresholding, drawing tools, cropping, webcam capture, histograms) across different stacks taught me a lot about how UI and processing logic should be separated cleanly, regardless of framework. I also learned how to package a Python app into a standalone executable using PyInstaller, so it can run without anyone needing to open a terminal.

Biggest challenge?

Interestingly, my biggest challenge this week had nothing to do with Computer Vision itself — it was environment-related. My machine had Python 3.14 installed, and I quickly found out that most of the libraries this project depends on (OpenCV, PyQt5, and others) were not yet compatible with that version. Instead of being a coding problem, it turned into a dependency and tooling problem, and it costed me real time.

How did you solve it?

I resolved it by removing Python 3.14 entirely and reinstalling a properly supported version through Homebrew, which gave me a clean, correctly linked Python installation that all the required libraries could actually install against. This experience taught me an important lesson early in the fellowship: the newest version of a tool is not

always the right version for a project, and checking library compatibility before setting up an environment can save a lot of wasted time. It also made me more comfortable diagnosing environment issues methodically instead of guessing, which is a skill I expect to rely on throughout this program.

Goals for Week 2

My main goal for Week 2 is to move beyond toolkit-style projects and start working on an actual real-time Computer Vision project something that processes a live video or webcam feed rather than static images, and applies detection and tracking in real time. I want to build on the pipeline knowledge and engineering habits from this week and put them to use on something closer to a real deployed CV application.